

IBVAP — Database Architecture & MongoDB Atlas Integration

06. Multi-Tier Persistence & Document Schema

1. MongoDB Configuration & Connection Specs

- **Database System:** MongoDB Atlas (Cloud NoSQL DB)
- **Database Name:** **IBVAP** (Configured in **src/app.py** Line 98)
- **Collection Name:** **intrusion_events** (Configured in **src/app.py** Line 99)
- **Driver / Library:** PyMongo (**pymongo.MongoClient**) + **certifi** (TLS CA Certificate verification)
- **Connection Mechanism:** Reads URI dynamically from environment variables (**MONGODB_URI**, **MONGO_URI**, **MONGO_URL**) or Streamlit secrets (**st.secrets["MONGODB_URI"]**).
- **Timeout Configuration:** **serverSelectionTimeoutMS=10000** (10 seconds)
- **API Version:** **ServerApi(version="1")**

2. Standard MongoDB Document Schema

Below is an exact example MongoDB document produced by **save_events_to_mongodb()** in **src/app.py**:

```
{
  "_id": {
    "$oid": "66dba12f8a4f9b2c3d4e5f60"
  },
  "event_id": "EVT-001",
  "event_type": "Zone Entry",
  "person_track_id": 2,
  "frame": 30,
  "timestamp": "00:01.20",
  "risk": "HIGH",
  "evidence_file": "https://images.pexels.com/photos/13530045/pexels-photo-13530045.jpeg?auto=compress&cs=tinysrgb&h=650&w=940",
  "status": "Confirmed",
  "detection_method": "Restricted Zone Entry",
  "created_at": "2026-09-06T10:33:44.368118"
}
```

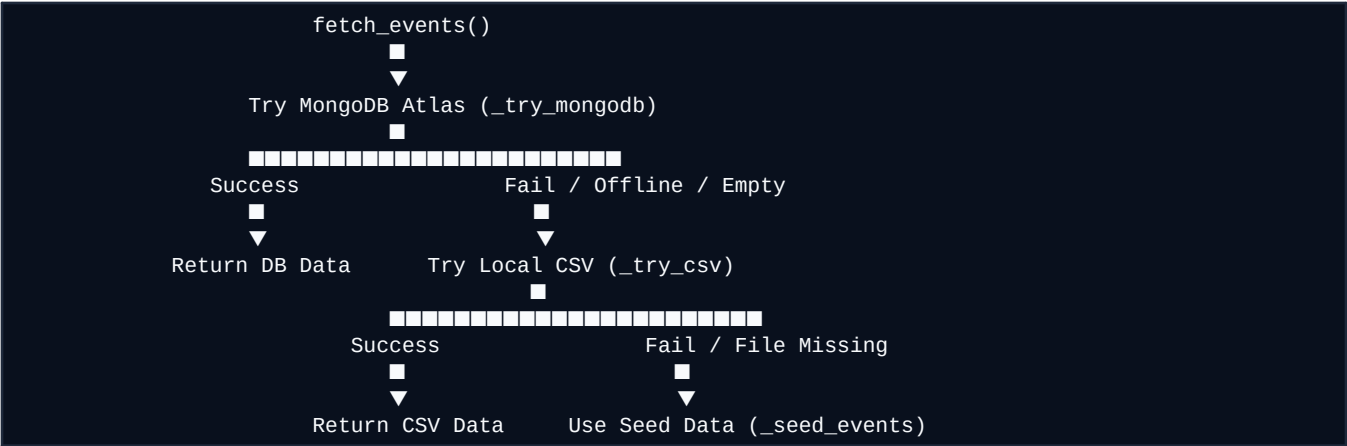
3. Field Descriptions

Field Name	Type	Description
event_id	String	Primary business key (e.g. "EVT-001"). Used for upsert queries.

event_type	String	Type of security event (e.g. "Zone Entry").
person_track_id	Integer	ByteTrack numeric session identifier (e.g. 2).
frame	Integer	Video frame number where breach was confirmed (e.g. 30).
timestamp	String	Video timestamp string in MM:SS.ms format (e.g. "00:01.20").
risk	String	Threat level classification ("HIGH", "MEDIUM", "LOW").
evidence_file	String / Null	File path or HTTPS URL of captured snapshot.
status	String	Operational status ("Confirmed", "Pending", "Resolved").
detection_method	String	Detection trigger rule ("Restricted Zone Entry").
created_at	String (ISO 8601)	System generation timestamp.

4. Multi-Tier Fallback Strategy & Resilience

To guarantee 100% uptime during SIH demonstrations or cloud deployments where MongoDB Atlas may be unreachable due to network restrictions, IBVAP implements an automatic 3-tier fallback loader in `src/app.py` (`fetch_events()`):



- **Tier 1 (Cloud DB):** `_try_mongodb()` attempts to ping MongoDB Atlas. If successful, loads live documents sorted by `event_id`.
- **Tier 2 (Local CSV):** `_try_csv()` parses `output/intrusion_log.csv` if MongoDB is down or unconfigured. Automatically syncs local CSV data up to MongoDB when connection restores.

- **Tier 3 (Seed Data):** `_seed_events()` returns hardcoded fallback prototype events if both MongoDB and local CSV are missing.
-

5. Why MongoDB instead of SQL/CSV?

- **Schemaless Flexibility:** Border surveillance sensors and multi-camera models frequently add metadata (e.g., thermal signatures, GPS coordinates, camera IDs, drone identifiers). MongoDB handles JSON field additions without database migration scripts.
- **High Write Throughput:** Document databases handle bursts of parallel edge events from multiple camera streams cleanly.
- **JSON Native Integration:** Direct 1-to-1 mapping with web API endpoints, Streamlit session state, and JSON audit ledgers.